Mishkal Tahor - User Manual

Version V11 (beta) · Mishkal by Lovy Elias · Nov 7, 2026

Welcome

Thank you for testing Mishkal Tahor. Tahor is three EQs in one window: **Reso** tames harsh resonances, **Tahor** is a clean 8-band surgical EQ, and **Character** adds the colour of vintage-style hardware EQs. It is part of the Mishkal plugin collection by Lovy Elias.

This manual covers version **V11 (beta)**. Tahor runs as VST3, AU and AAX on Mac, and VST3 on Windows.

The sound always flows in this order:

Input > Reso > Tahor > Character > Output

So Reso cleans the sound first, Tahor shapes it, and Character colours the result.

Quick start

1. **Insert Tahor** on a track (a vocal is a great first test) and press play. The analyzer only appears while audio is playing.
2. **Set your level.** Click **AUTO** next to IN in the bottom bar. Tahor listens for about 3 seconds and sets the input to sit around -18 dBFS RMS.
3. **Pick a starting point.** Click the star next to the preset name to open the preset browser, and choose a preset from a folder such as Vocals or Drums.
4. **Choose a section.** Use the RESO / TAHOR / CHARACTER selector at the top (or press 1, 2 or 3). Double-click the screen in Reso or Tahor to add a band, then drag it.
5. **Compare fairly.** Turn on **MATCH** and use **BYPASS** or **A > B** to hear whether your changes really make it better, not just louder.

The window at a glance

The window has four parts: the header along the top, the big screen in the middle, a meter on each side, and the bottom bar. Everything here works the same in all three sections.

Header (top row, left to right)

Control	What it does
Tahor name	Shows which plugin you are in.

Control	What it does
Undo / Redo	Steps back or forward through your changes.
History	Opens a list of your recent changes by name, so you can jump back several steps at once.
Preset name with < >	Steps through presets one at a time.
Star	Opens the preset browser.
Preset menu (small arrow)	Save, Save As (with a tag) and Browse presets.
A > B	Compares two versions of your settings.
MATCH	Matches output loudness to the input, so comparisons are fair.
BYPASS	Switches the whole plugin off so you can hear the original.
RESO / TAHOR / CHARACTER selector	Chooses the section you are editing. The arrows either side flip through the three. The power icon switches the current section on or off; a switched-off section goes dull.
AUTO GAIN	Estimates how much louder or quieter your EQ makes the sound and corrects it at the output. Shows the correction in dB.
Mishkal logo	The Mishkal collection badge.

The screen

The large middle area shows your EQ curves over a live analyzer. The darker layer is the sound **before** Tahor (pre) and the brighter layer is the sound **after** it (post). The analyzer only appears while audio is playing.

The square button in the top-right corner of the screen switches to full screen.

Meters

The input meter runs down the left edge and the output meter down the right edge. At the bottom of each you will see two readouts: **PK** (the peak, held) and **AVG** (the average level).

Bottom bar (left to right)

Control	What it does
IN	Input gain.
AUTO	Listens for about 3 seconds and sets the input so it sits around -18 dBFS RMS.
Phase mode	**Natural Phase** (low latency), **Linear Phase**, or **Linear Phase HQ**.
Oversampling	Off up to 32x. Mainly helps the Character section's colour sound smooth.
Render	The oversampling used when you bounce or export.
Analyzer	Pre + Post, Post, Pre or Off.
Range	The display scale of the screen: ±3, ±6, ±12 or ±30 dB. It changes the view, not the sound.
Size	Window size, from Mini to Extra Large.
SC Ext	Uses the external sidechain input from your DAW.
OUT	Output gain.

Reso: the resonance suppressor

Reso finds harsh, ringing frequencies and dips them only at the moments they ring. The rest of the time it leaves the sound alone. Use it for harsh vocals, boxy rooms, fizzy guitars and sibilance.



Figure 1. The Reso section, with the Depth panel on the left and a band panel open.

Reading the screen

- **The red curve at the top** shows what Reso is removing, in real time. It hangs down from the top of the screen; the deeper a dip, the more Reso is cutting there.
- **The red edges** (vertical lines at the left and right) are the focus range. Drag them in to make Reso work only between those frequencies.
- **The dots** are shaping bands. They tell Reso to push harder (more) or relax (less) at a frequency. They do not boost or cut the sound directly.

Shaping bands

Double-click the screen to add a shaping band. Shapes available: Bell, Bandpass, Low shelf, High shelf, Tilt, Notch, Low cut and High cut.

Touch a dot to open its panel:

- **Amount**: how much more (+) or less (-) Reso works at that frequency.
- **Width**: how wide the band is (shown as Q).
- **Shape**: the dropdown at the bottom of the panel.
- **ST / L / R**: which side the band works on: both (stereo), left only or right only.
- **S**: solo, to hear just that area.

The bubble above the dot shows its number, shape, frequency and amount, for example "2 · Bell · 970 Hz · +3 dB".

Left panel

The main Reso controls live in the panel on the left. Click the small tab on its edge to slide it in or out.

Control	What it does
Soft / Hard	Gentle, smooth suppression or a firmer, more obvious one.
Low latency	Reduces delay, useful while recording or playing live.
Depth	How strongly Reso cuts overall. Start here.
Detail	How narrow and precise the cuts are. Higher values catch smaller, sharper resonances.
Attack	How quickly Reso reacts when a resonance appears.
Release	How quickly Reso lets go after it stops.
Max cut	The most Reso is ever allowed to cut, for example -12 dB.
Mix	Blends the processed and original sound.
Trim	Output level of the Reso section.
Reduction	A meter showing how much Reso is cutting right now.
Stereo / Mid / Side	Process the whole signal, only the middle, or only the sides.
Delta	Lets you hear only what Reso is removing. Great for checking you are not cutting too much.
Sidechain	Lets another signal decide when Reso works.

Advanced panel

Click the **ADV** tab on the right edge to open the Advanced panel.



Figure 2. Reso with the Advanced panel open on the right.

Control	What it does
Detail tilt	Leans Detail toward the lows or the highs.
Attack tilt	Leans Attack toward the lows or the highs.
Release tilt	Leans Release toward the lows or the highs.
Stereo link (Link)	How much the left and right channels react together. 100% means they move as one.
Phase: Minimum / Linear	Linear keeps parallel and Mid/Side processing working cleanly.

Tip: turn on Delta and raise Depth until you hear only the harsh ringing in the Delta signal, then turn Delta off.

Tahor: the surgical EQ

Tahor is a clean, precise EQ with up to 8 bands. Use it to remove mud, notch out a ring, cut rumble or gently lift a frequency.

Working with bands

- **Add a band:** double-click the screen.
- **Move a band:** drag its dot left or right to change the frequency, up or down to change the gain.
- **Change the width (Q):** roll the mouse wheel over the dot.
- **Read the values:** the bubble above the dot you touch shows its shape, frequency and gain live, for example "Bell · 1 kHz · +4 dB".

The band panel

Touch a dot to open its panel at the bottom of the screen.

- **Shape:** Bell, Low shelf, High shelf, Low cut, High cut or Notch. The cuts have slopes from 6 to 48 dB per octave.
- **ST / M / S / L / R:** where the band works: stereo, mid, side, left or right.
- **Knobs:** frequency, gain and Q.
- **OPTIONS menu:** sets the band's mode to **EQ** or **Dynamic**, and turns on **Sidechain**.
- **S:** solo the band.

Dynamic bands

A Dynamic band only moves when the sound crosses its threshold, so it cuts a harsh note only while it plays. In Dynamic view the panel shows:

Control	What it does
Threshold	The level where the band starts to act.
Range	The most the band can move, for example -8 dB.
Attack	How fast it reacts.
Release	How fast it lets go.
Q	The width of the band.
Live meter	How many dB the band is pulling right now.

On screen, three shapes show what is happening:

- **Solid curve:** the gain you set.
- **Shaded curve:** the reduction happening right now.
- **Dashed outline:** the full range the band may move.



Figure 3. A dynamic band in Tahor, with the frequency bubble and the band panel in Dynamic view.

Splitting a band into left and right

Press **L** or **R** in the band panel. Tahor makes a partner band on the other side with the same settings. From then on, each is independent, so you can set the left and right differently.

- The left curve is **blue** and the right curve is **pink**.
- The dots are tagged **L** and **R**.
- Press **ST** to join them again.

Solo

Press **S** in the band panel (or the S key). The soloed band's range lights up as a column on the screen, as wide as its Q, and the rest of the screen dims. You hear only that part of the sound, which makes it easy to find a problem frequency.



Figure 4. A band split into left (blue) and right (pink), with a band soloed: the lit column.

Character: vintage-style colour

Character adds the tone of classic hardware EQs. You can chain up to 4 units, each inspired by a famous piece of studio gear and given its own LE name.

Adding units

Character starts empty. Nothing is processing yet; LE-EP1 is shown only as a preview.

1. Click **+ Add EQ** to open the gear gallery.
2. Click a faceplate to add that unit.
3. Repeat for up to 4 units in a chain.



Figure 5. The + Add EQ gear gallery, showing all 16 units with their one-word hints.

The units in your chain appear as chips on the left of the screen, in the order the sound passes through them. Drag a chip to reorder, or click its **x** to remove it.

The faceplate

The hardware faceplate at the bottom of the screen shows the selected unit. Hover over it and arrows appear either side, so you can preview other units. A one-word hint above the faceplate (such as "Weight") tells you what the unit is best at.

The larger two-rack-unit gear (LE-EP1, LE-MP2, LE-Esh, LE-Even and LE-Kavod) is laid out like the originals: big gain knobs above a row of frequency selectors.

Reading the curves

- Each unit draws its **own coloured curve**.
- The **bright line** is the combined sound of the whole chain.
- The legend at the top left shows which colour belongs to which unit.

Nodes

The numbered dots on the curve drag just like Tahor's bands, but they snap to the steps the hardware offers. When a unit boosts and cuts the same frequency (the Pultec-style trick), the two dots sit slightly apart, joined by a line.

Touch a dot to open its node panel:

- **Knobs** for that node.
- **ST / L / R:** split the whole unit into separate left and right settings.
- **S:** solo the node's range.

Colour and THD

The top row has the colour switch, **Clean / Subtle / Warm**, and the **THD** slider, which sets how much harmonic colour is added. 50% is the normal amount. For the smoothest colour, raise **Oversampling** in the bottom bar.



Figure 6. Character with LE-EP1 and LE-Esh in the chain and the LE-EP1 faceplate on screen.

The 16 units

All units are inspired by classic hardware. They are Mishkal's own designs, not the original brands.

Unit	Inspired by	Hint
LE-EP1	Pultec EQP-1A-style tube program EQ	Weight
LE-ME5	Pultec MEQ-5-style midrange EQ	Mids

Unit	Inspired by	Hint
LE-E4K	British E-series console EQ	Bite
LE-N73	1073-style preamp EQ	Warmth
LE-N81	Class A 81-style EQ	Girth
LE-550B	550B-style stepped EQ	Punch
LE-K9	9000K-style console EQ	Clean
LE-H69	Helios 69-style EQ	Grit
LE-SP1	Air EQ	Air
LE-M4	EQ4-style broad bands with Air	Open
LE-MP2	Massive Passive-style passive tube EQ	Gloss
LE-Esh	HG-Q-style tube EQ	Fire
LE-Shemen	Sontec MES-432C-style mastering parametric	Silk
LE-Kav	SPL PQ-style precision mastering EQ	Precision
LE-Even	Passeq-style passive EQ	Foundation
LE-Kavod	EAR 825Q-style passive tube mastering EQ	Elegance

Presets

Tahor comes with 69 factory presets, sorted into folders. They are starting points: load one, then adjust it to your sound.

Browsing

- Click the **star** next to the preset name to open the browser. Click a folder chip at the top (All, Vocals, Drums and so on) to filter the list, then click a preset to load it.
- Use the **< >** arrows beside the preset name to step through presets without opening the browser.
- Press **Esc** to close the browser.

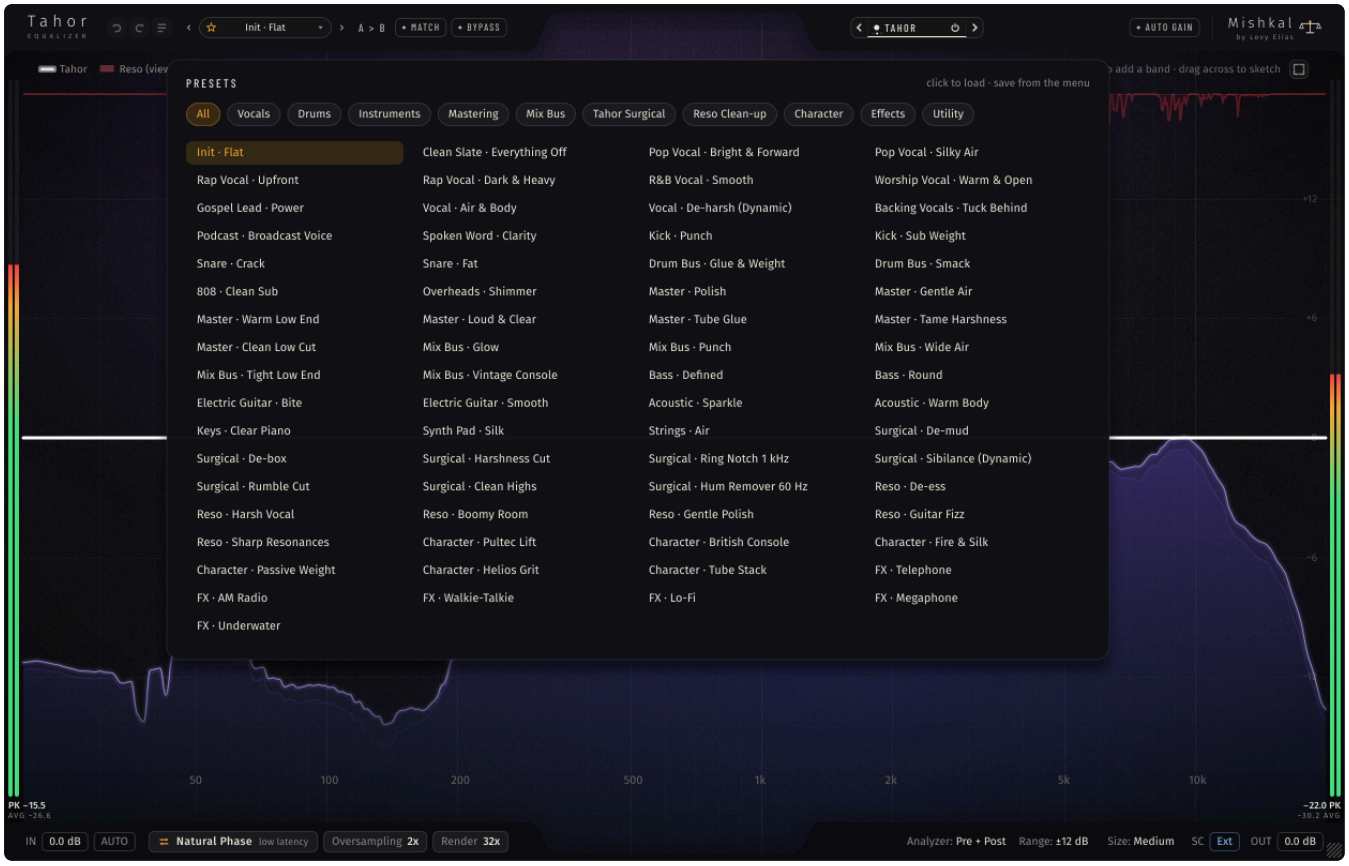


Figure 7. The preset browser with its folder chips.

Factory folders

Folder	What's inside
Vocals	Pop, rap, R&B, worship, gospel, podcast, spoken word, backing vocals
Drums	Kick, snare, drum bus, 808, overheads
Instruments	Bass, guitar, acoustic, keys, pads, strings
Mastering	Polish, air, low end, loudness and tone for the master bus
Mix Bus	Glow, punch, width and low-end control for the mix bus
Tahor Surgical	De-mud, de-box, harshness, ring notch, sibilance, rumble, hum remover
Reso Clean-up	De-ess, harsh vocal, boomy room and more
Character	Colour chains built from the vintage-style units

Folder	What's inside
Effects	Telephone, AM radio, walkie-talkie, lo-fi, megaphone, underwater
Utility	Init (flat start) and Clean Slate (everything off)

Saving your own

Open the preset menu (the small arrow beside the preset name):

- **Save** updates the current preset.
- **Save As** saves a new preset with a tag. The tag becomes its folder in the browser. If you don't choose one, it goes under **Mine**.
- **Browse presets** opens the browser.

Your presets are stored on Mac in:

`~/Library/Audio/Presets/Mishkal Audio/Tahor`

Metering and gain

Louder almost always sounds "better", so Tahor gives you three ways to keep levels honest.

- **IN AUTO** (bottom bar): listens for about 3 seconds and sets the input gain so the sound sits around -18 dBFS RMS.
- **AUTO GAIN** (header): estimates how much louder or quieter your EQ settings make the sound, and corrects it at the output. The correction is shown in dB next to the button.
- **MATCH** (header): matches the output loudness to the input, so when you compare with BYPASS or A > B you hear the change in tone, not a change in volume.

Meters and readouts

- **Input meter:** the left edge of the window.
- **Output meter:** the right edge of the window.
- **PK:** the highest peak, held so you can read it.
- **AVG:** the average level, which is closer to how loud it feels.

Compare the input and output readouts to see how much your settings change the level. Use **OUT** in the bottom bar for the final level.

Tip: in Reso, the Reduction meter in the left panel shows how much is being cut. In a Dynamic band in Tahor, the small live meter shows how many dB the band is pulling.

Keyboard and mouse shortcuts

Shortcut	What it does
1 / 2 / 3	Switch to Reso / Tahor / Character
Cmd-Z	Undo
Shift-Cmd-Z	Redo
Delete or Backspace	Remove the selected band (Tahor or Reso)
Left / Right arrow	Nudge the selected band's frequency
Up / Down arrow	Nudge the selected band's gain
Shift + arrow	Nudge in finer steps
S	Solo the selected band
B	Bypass
M	Match
Esc	Close the gear gallery or the preset browser
Double-click or Alt-click a knob or dot	Reset it
Mouse wheel over a dot	Change its Q (width)
Double-click the screen	Add a band (Reso or Tahor)

Installing

Requirements

	Mac	Windows
System	macOS 12 or later	Windows 10 or 11, 64-bit
Processor	Apple Silicon or Intel	64-bit
Formats	VST3, AU, AAX	VST3

Mac

1. Run the installer **.pkg**.
2. This beta is not signed yet, so macOS may say it can't be opened. If it does, go to **System Settings > Privacy & Security** and click **Open Anyway**. (Or right-click the installer and choose **Open**.)
3. Restart your DAW, or rescan your plug-ins.

Windows

1. Run the installer **.exe**.
2. If SmartScreen shows a warning, click **More info**, then **Run anyway**.
3. The VST3 is installed to `C:\Program Files\Common Files\VST3`. Restart your DAW, or rescan your plug-ins.

Feedback

This is a beta, and your ears help shape the final version. Please tell Lovy:

- What you used Tahor on (vocal, drums, mix bus, master and so on).
- Which DAW and computer you used.
- Anything that looked wrong, sounded wrong, or was confusing.
- What you loved, and what you wish it did.

Screenshots and short audio examples are especially helpful. Thank you for testing.